

Problem

Sketch the Bode plots for

$$G(s) = \frac{s}{(s+2)^2(s+1)}, \quad s = j\omega$$

Step-by-step solution

Step 1 of 4

Consider the transfer function, $G(s) = \frac{s}{(s+2)^2(s+1)}$.

Substitute $j\omega$ for s .

$$\begin{aligned} G(j\omega) &= \frac{j\omega}{(j\omega+2)^2(j\omega+1)} \\ &= \frac{j\omega}{4\left(\frac{j\omega}{2}+1\right)^2(j\omega+1)} \\ &= \frac{\left(\frac{1}{4}\right)j\omega}{(1+j\omega)\left(1+\frac{j\omega}{2}\right)^2} \end{aligned}$$

Step 2 of 4

Determine the magnitude of the transfer function.

$$\begin{aligned} G_{dB} &= 20 \log_{10} \left(\frac{1}{4} \right) + 20 \log_{10} |j\omega| - 20 \log_{10} |1+j\omega| - 20 \log_{10} \left| 1 + \frac{j\omega}{2} \right|^2 \\ &= 20 \log_{10} \left(\frac{1}{4} \right) + 0 - 20 \log_{10} |1+j\omega| - 40 \log_{10} \left| 1 + \frac{j\omega}{2} \right| \\ &= -12 - 20 \log_{10} |1+j\omega| - 40 \log_{10} \left| 1 + \frac{j\omega}{2} \right| \end{aligned}$$

Determine the phase of the transfer function.

$$\phi = 90^\circ - \tan^{-1} \omega - 2 \tan^{-1} \frac{\omega}{2}$$

Step 3 of 4

The following is the magnitude plot for the transfer function:

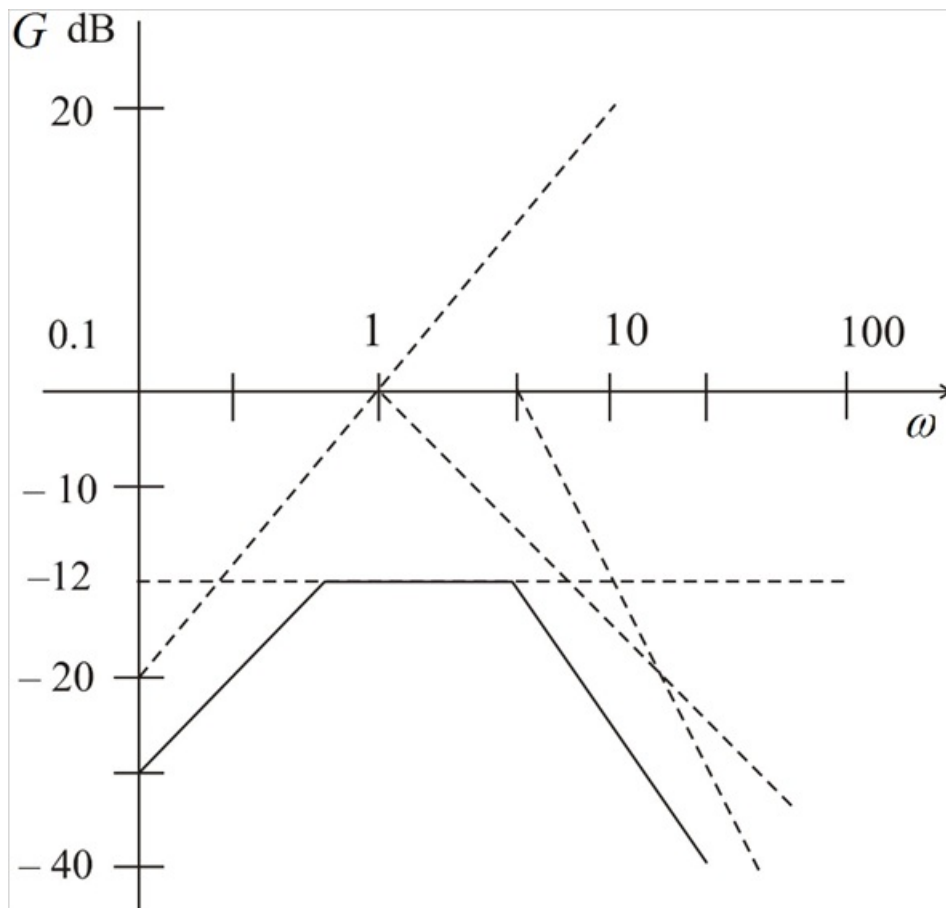


Figure 1

Step 4 of 4

The following is the phase plot for the transfer function:

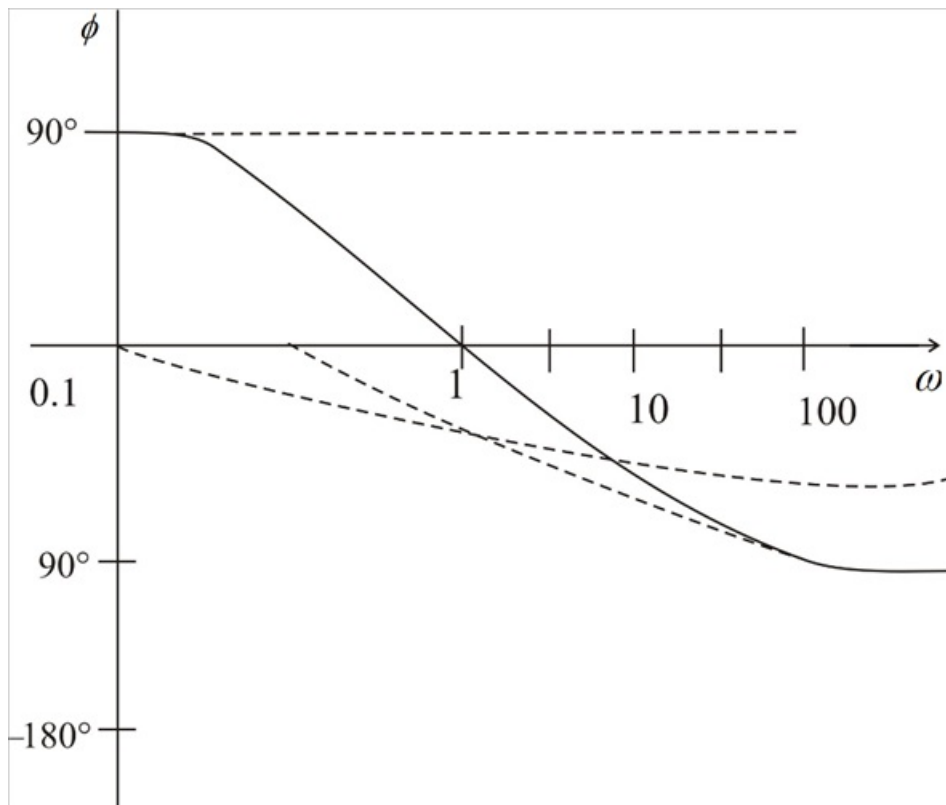


Figure 2

Thus, the Bode plot is drawn.